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Toward a sustainable and accessible marimba: Acoustic screening of commercially available materials as alternatives to Honduran rosewood

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The marimba is a tuned idiophone whose bars are often made from Honduran rosewood, a scarce and expensive material that limits access for students, schools, and aspiring percussionists. This study investigates commercially available American woods, plywoods, and printed plastics as cost effective and sustainable alternatives for marimba bars, with emphasis on preserving properties related to musical performance. Prior work has largely characterized existing marimba materials or evaluated individual substitute species, leaving a gap for comparative screening of accessible alternatives. Tested materials were evaluated by density, Young's modulus, and loss factor, which together influence durability, pitch behavior, playing feel, and resonance. No tested material besides the rosewood reference and padauk met the original screening criteria, however, hickory and cherry demonstrated enough promise to justify further study. Hickory was the strongest candidate as its density, Young's modulus, and stiffness to density behavior were comparatively close to rosewood. Cherry was less dense but retained encouraging stiffness to density and damping behavior. These results provide foundational data for developing more accessible and environmentally sustainable practice marimba bars.

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INTRODUCTION

The marimba is a tuned idiophone: it generates sound through the vibration of its bars, rather than through vibrating strings or moving air. Its roots are commonly traced either to 14th-century Southeast Asia or to mid-16th-century West Africa, with later development in the Americas through the African slave trade [1]. The marimba is now culturally significant across the world, especially across parts of Latin America [2]. The modern concert marimba was further shaped in 1890, when J. C. Deagan and U. G. Leedy added tube-shaped resonators under the bars, allowing the instrument to project over large ensembles such as orchestras [2].

A central challenge for marimbists and instrument builders is the sourcing of bar wood. Marimbas are often built using trees from the Dalbergia genus (rosewoods), especially Honduran rosewood (Dalbergia stevensonii) [3]. The wood is valued for its color and acoustic response; however, excessive logging, slow growth cycles, deforestation, and trafficking make responsible sourcing difficult [3]. Despite these concerns, rosewood remains the primary wood used for high-quality marimba bars [2]. Its price contributes to access barriers: standard 4.3-octave rosewood marimbas from reputable manufacturers cost approximately $7,000 and above [4]-[6]. Students who do have access to school instruments may still face limited practice time as multiple percussionists often share a small number of instruments. This disparity can hinder the progress of percussion students relative to non-percussion peers who can practice more easily at home.

The scholarly literature directly addressing marimba bar materials is limited. Existing work generally evaluates the acoustic and mechanical properties of known viable wood types such as padauk, jaratoba, and merbau [7], or tests a single substitute such as Kayu Malam [8] or European mountain ash [9]. Broader work on wood acoustics, including Wegst's study of wood for musical instruments [10] and Bucur's Acoustics of Wood [11], provides valuable context for musical instrument materials. Bucur identifies several potential substitute woods, but most are already in limited use, raise conservation concerns, or lack damping data needed for full mallet-instrument assessment [11]-[14]. This leaves a gap around commercially available American materials and around low-cost screening methods that can identify candidates before complete tuned bars are fabricated.

Current alternatives to rosewood also face limitations. Padauk, a sometimes-used natural substitute from West Africa, is less expensive than rosewood but still costly: Marimba One's padauk educational marimba is listed at $5,799, compared with $7,699 for its rosewood Motif model [4], [15]. Padauk also creates manufacturing difficulty because dense tropical woods can wear tools quickly [16]. Several companies, including Adams, Majestic, and Yamaha, have developed proprietary synthetic materials [5], [6], [17], [18]. These options remain expensive due to complex production methods, limited market size, research and development costs, and trade secrets. Electronic mallet controllers, such as the Pearl MalletSTATION EM1, are cheaper but are not acoustic substitutes for practicing resonance, rolling, and physical bar response [19].

Three variables determine whether an alternative material has potential for marimba bars: density (ρ), Young's modulus (E), and loss factor (η) [20]. Density is used as a preliminary proxy for durability as lower-density bars are more susceptible to cracking under repeated strikes. Young's modulus measures stiffness and plays a role in determining both playing feel and the formation of overtones after a strike. The Young's-modulus-to-density ratio (E/ρ) is especially important as it determines the fundamental pitch of a bar with a given size and shape [20]. Materials with the same E/ρ ratio will produce the same pitch when crafted into bars of identical geometry. The loss factor, also called damping, represents internal friction and determines how quickly sound decays after a strike [20]. Rosewood marimbas are valued partly because this decay is slow, supporting resonance and performance techniques such as rolls [10].

Because rosewood is expensive and difficult to source responsibly, alternative materials could increase access and improve sustainability. The research question is intentionally framed around commercially available materials as an excellent technical substitute fails to address the access issue if the material is difficult to supply. This framing also separates two questions that are sometimes combined: whether a material can sound exactly like Honduran rosewood, and whether it can create a durable, affordable, pedagogically useful marimba bar. The present work focuses on the latter while using rosewood as the acoustic reference point. Candidate materials were compared with Honduran rosewood using two criteria: an E/ρ ratio within ±20% of rosewood's 0.0168 $(GPa*m^3)/kg$ value and a loss factor within ±75% of rosewood's 0.0101 value. The E/ρ tolerance was chosen as a practical limit for pitch producing geometry so that practice on an alternative bar would transfer to a rosewood marimba without requiring drastically different dimensions. The damping margin was chosen because it approximates the difference between rosewood and padauk; since padauk is used in marimbas, materials with damping near that range may warrant further study. Because damping is difficult to infer from general wood-property tables, measuring η directly was a central part of the study rather than a secondary check.

METHODS

Materials and density measurements

Two processes were required to measure the primary variables: direct measurement for density and vibrational analysis for Young's modulus and damping. The study used three categories of materials: solid woods, plywoods, and plastics. Solid woods were selected to represent wood types commonly found in American woodworking shops. Plywoods were included due to their low cost and purported environmental benefits [21]. One sample of each material was tested (n = 1), except for cherry, which had two samples (n = 2).

Specifically, the materials used were Hickory (*Carya* spp.), white ash (*Fraxinus americana*), padauk (*Pterocarpus soyauxii*), red oak (*Quercus rubra*), cherry (*Prunus serotina*), fir (*Pseudotsuga menziesii*), PLA (Bambu Lab PLA Basic, blue, 1.75 mm), PLA+ (Overture PLA Professional, blue, 1.75 mm), walnut (*Juglans nigra*), maple (*Acer saccharum*), white oak (*Quercus alba*), poplar (*Liriodendron tulipifera*), pine (*Pinus* spp.), 5-ply pine plywood (southern yellow pine, *Pinus* spp.; Plytanium 19/32 in. BCX sanded plywood, Georgia-Pacific, Lowe's), 4-ply pine plywood (southern yellow pine, *Pinus* spp.; Plytanium 15/32 in. rated pine plywood sheathing, Georgia-Pacific, Lowe's), 9-ply Baltic birch plywood (*Betula* spp.; Rockler Woodworking and Hardware), cherry 2 (*Prunus serotina*), bamboo plywood (*Phyllostachys edulis*; laminated from thin commercial sheet).

Due to Honduran rosewood's prohibitive cost, a sample was unavailable for testing. Its values were gathered from Culik and Danihelova [9]. To verify the success of the methodology, a padauk sample was included as a positive control. Padauk is also used in marimba production but still remains costly, meaning a more affordable alternative would still be valuable. Because padauk is already used in marimbas, finding that its properties approach rosewood's is not a new discovery. Two cherry samples from different trees were included to illustrate variation among samples of the same species. Because a piece of bamboo plywood thick enough for testing was unavailable, a long, thin piece was cut into sections and glued with Titebond wood glue to approximate a thicker bamboo-plywood sample. This bamboo result should therefore be interpreted as a rough approximation rather than as a definitive measurement of commercial bamboo plywood.

The two plastics tested were polylactic acid (PLA) and a toughened PLA (PLA+). PLA is a plastic frequently used in 3D printing [22]. The PLA samples were printed with 99.9% infill and 20,000 mm/s² acceleration so that measured density and Young's modulus would reflect the materials' inherent characteristics rather than the structural effects of a low-infill print.

The tested material set included hardwoods, softwoods, plywoods, and plastics, as shown in Fig. 1. Hardwoods and softwoods were common commercially available American materials; plywoods were included for cost and sustainability reasons; and plastics were included to test whether 3D-printable polymers could approximate marimba-bar properties.

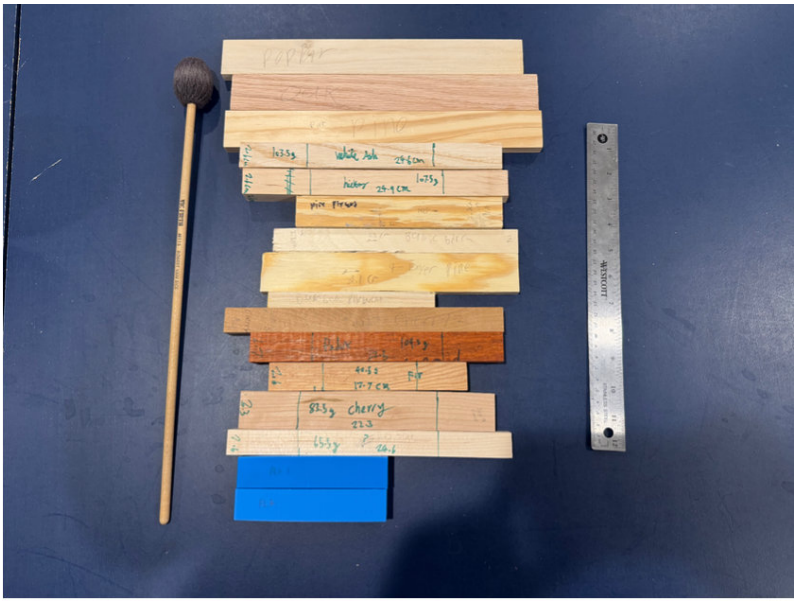


Figure 1. All tested material types besides walnut and maple. This includes hardwoods, plywoods, and plastics. In order: poplar, white oak, pine, white ash, hickory, 5-ply pine plywood, 9-ply Baltic birch, 4-ply pine plywood, bamboo plywood, cherry 1, padauk, fir, cherry 2, red oak, PLA, PLA+

Density required mass and volume. The mass of each sample was determined using a Pohl-Schmitt food scale with ±0.5 g precision. Volume was calculated using a ruler with ±1 mm precision by measuring height, length, and width, and density was then determined by dividing mass by volume. Dimensional analysis was used instead of water displacement because water would alter the humidity of the samples and yield non-representative results. While a caliper and more precise scale would have produced more granular data, the relatively large sample dimensions made millimeter-level precision acceptable for an exploratory comparison.

Solid-wood samples were prepared with the grain running along the length of the sample. Plywood samples were cut along the length of the sheet, so the face grain of each plywood is expected to run along the sample length (not verified).

Audio acquisition and processing

To determine Young's modulus and damping, an audio sample of each material's natural resonance was required. Each block was suspended on two twine strings placed near the first bending-mode nodal region, approximately one-quarter of the sample length, to reduce support interference with natural resonance [20]. This method was used instead of laser vibrometry as laser vibrometry requires specialized laboratory equipment and was outside the project's cost and availability constraints [23]. Sound samples were taken once per material.

To reduce interference and ensure that the microphone captured emitted sound, the twine was attached to a rectangular box with the top and one side removed. The microphone was placed slightly inside the shorter open side of the box, and the recording setup shown in Fig. 2 was used. Each material sample was tapped with the wooden end of a Vic Firth Robert Van Sice M115 mallet, and the resulting sound was recorded with a Sudotack ST-810 microphone. The approach was intended to isolate each sample's inherent vibrational frequency while preserving a setup that could be replicated without specialized laboratory equipment.

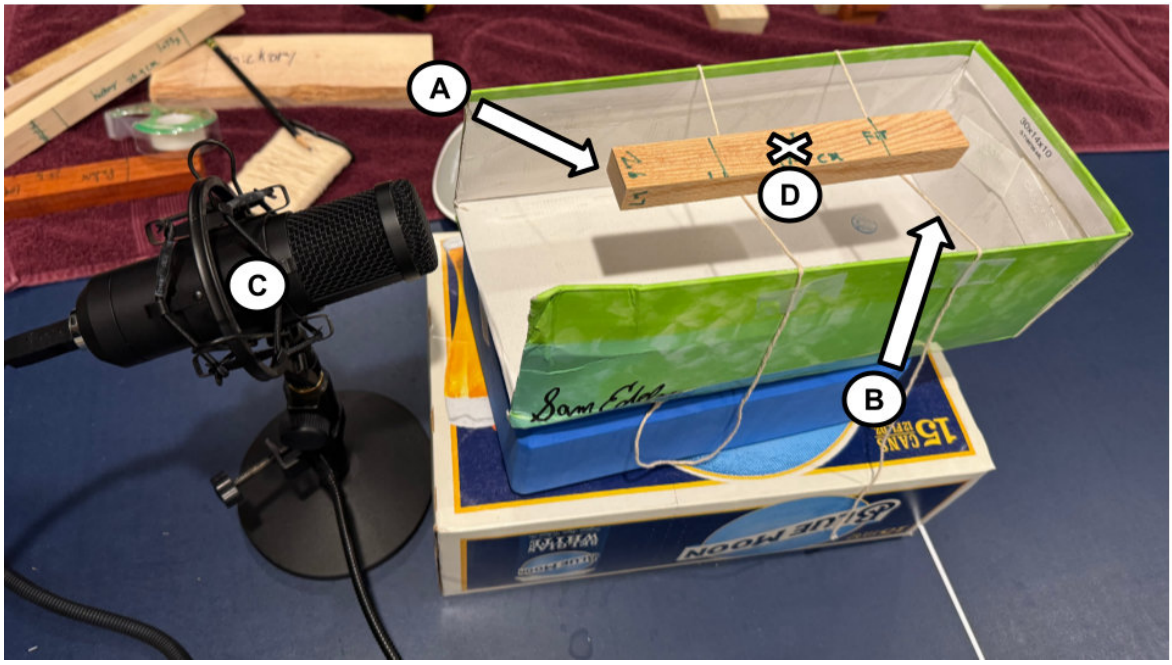


Figure 2. Suspended-sample recording setup used for tap-response measurements: A – sample; B – twine support at nodal line; C – microphone; D – strike point

Each sound sample was processed in Audacity 3.7.3 using the Fast Fourier Transform (FFT) "plot spectrum" feature. The sampling rate was 44.1 kHz, and the FFT size was 1024 with a Hann window function. The fundamental frequency was identified as the lowest-frequency prominent flexural resonance peak, excluding low-amplitude background noise and non-resonant low-frequency artifacts. After combining the fundamental frequency with the density and dimensions of the material sample, Young's modulus was calculated using the equation in Euphonics, section 10.3.1 [20]:

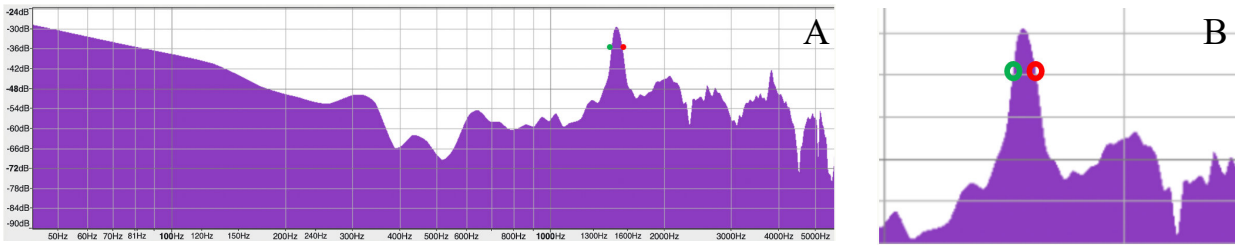


Figure 3. A — FFT plot spectrum used to identify the fundamental flexural resonance with f_0 at the peak, and at -3dB from the peak, f_- in the green circle and f_+ in the red circle; B — zoomed in image of plot spectrum

$$E = 0.946\rho\frac{f_0^2 L^4}{h^2} \quad (1)$$

In Eq. (1), ρ represents density, f_0 represents the fundamental flexural frequency, L is sample length, h is the thickness in the bending direction, and E is Young's modulus. This method is used in related literature on acoustic and mechanical properties of wood [24], [20], [25].

The loss factor was obtained with the half-width bandwidth method shown in Fig. 3. Starting from the fundamental peak at f_0, the spectrum was followed in both directions until the sound intensity level decreased by 3 dB on each side. The upper and lower frequencies at which the spectrum fell 3 dB below the peak level were denoted $f+$ and $f-$.The loss factor was calculated as

$$\eta = \frac{f_+ - f_-}{f_{peak}} \quad (2)$$

A 3 dB reduction was used because decibels are logarithmic and a 3 dB change corresponds approximately to half the sound intensity. Audacity provides precise frequency readings, but its cursor tool is limited to whole-decibel values, introducing possible error in loss-factor calculations. Data were consolidated in Google Sheets. After density, Young's modulus, E/ρ, and damping were calculated, exported CSV data were visualized with pandas and matplotlib in Python.

The sample-preparation choices were intended to reduce avoidable variation without requiring a professional instrument shop. For the 3D-printed samples, high infill was used so that the sample behaved more like a solid material than a lightweight printed lattice. For the bamboo plywood approximation, glued sections were used because a thick sample was unavailable; this makes the bamboo result less definitive than the solid-wood results, but still useful for exploratory comparison. The two cherry samples were intentionally retained rather than averaged during data collection because wood is not a perfectly uniform industrial material. Their differences illustrate why future work needs multiple samples from each species before making manufacturing-level conclusions.

An accelerometer-based damping measurement was considered but not used. Attaching an accelerometer to the end of a material sample can provide useful vibration data, but the accelerometer's mass and attachment method can itself affect damping. Because damping was one of the primary variables in this study, adding a physical sensor to the vibrating sample risked interfering with the property being measured. The microphone method was therefore more appropriate for this study.

The study's processing pipeline was designed to preserve replicability. Each sample produced a mass, three-dimensional measurements, a fundamental frequency, two half-width bandwidth frequencies, a calculated density, a calculated Young's modulus, a calculated E/ρ ratio, and a calculated loss factor. Consolidating the data in a spreadsheet made it possible to check calculations repeatedly before exporting the data for plotting. This structure is important for future researchers because it separates directly measured quantities from derived quantities, making it easier to identify which measurements would benefit most from improved instrumentation.

RESULTS

Table 1: Dimensions, mass, and measured acoustic properties of candidate bar samples. Width (X), thickness (Y), and length(Z) are in meters. Density ρ was computed from measured mass and volume. Pitch is the measured fundamental frequency f_0. f2 and f1 are the lower and upper -3dB points. Young's modulus was estimated from f_0, ρ, length, and thickness. Damping estimated from f_0 and the -3dB points.

Wood	X (Width)	Y (Height)	Z (Length)	Volume (m^3)	Mass (kg)	Density (kg/m^3)	Pitch (Hz)	Young's modulus (Gpa)	f2 (-3dB right side)	f1 (-3dB left side)	Damping
Hickory	0.026	0.021	0.249	0.000136	0.108	791	1550	15.7	1563	1524	0.0252
White Ash	0.026	0.022	0.246	0.000141	0.104	736	1911	19.2	1954	1872	0.0429
Padauk	0.027	0.024	0.225	0.000146	0.110	751	1963	12.2	1979	1946	0.0168
Red Oak	0.026	0.017	0.246	0.000109	0.0655	602	1515	16.6	1543	1491	0.0343
Cherry	0.033	0.019	0.224	0.000140	0.0835	595	1489	8.70	1508	1470	0.0255
Fir	0.026	0.017	0.177	0.0000782	0.0405	518	2762	12.7	2807	2742	0.0235
PLA	0.025	0.025	0.126	0.0000788	0.0965	1230	2371	2.63	2424	2305	0.0502
PLA+	0.025	0.025	0.126	0.0000788	0.091	1160	2239	2.21	2287	2201	0.0384
Walnut	0.037	0.02	0.194	0.000144	0.0905	630	2224	10.4	2354	2222	0.0594
Maple	0.032	0.021	0.223	0.000150	0.0795	531	1866	9.80	1900	1828	0.0386
White Oak	0.038	0.013	0.3	0.000148	0.104	702	725	16.7	738	707	0.0428
Poplar	0.039	0.013	0.299	0.000152	0.072	475	697	10.3	716	674	0.0603
Pine	0.038	0.013	0.316	0.000156	0.072	461	650	10.9	688	612	0.117
5-Layer Pine	0.029	0.014	0.194	0.0000788	0.0595	755	1616	13.5	1660	1567	0.0575
4-layer Pine	0.037	0.012	0.241	0.000107	0.0645	603	807	8.70	848	767	0.100
9-layer Baltic birch	0.023	0.013	0.23	0.0000688	0.0405	589	918	7.77	934	895	0.0425
Cherry 2	0.023	0.021	0.251	0.000121	0.0675	557	1402	9.32	1423	1383	0.0285
Bamboo plywood	0.019	0.016	0.154	0.0000468	0.027	223	1829	1.55	1886	1780	0.0580
Rosewood	-	-	-	-	-	950	-	16	-	-	0.0101

Density and Young's modulus

The density results are summarized in Fig. 4. Density is significant because it relates directly to durability in mallet percussion, where the material is repeatedly struck to create sound. The two PLA-based materials had the highest densities by a wide margin, likely because the samples were printed at 99.9% infill while natural wood materials contain air pockets. Density alone was not used as a final pass/fail criterion. Instead, Fig. 4 groups materials descriptively using the measured walnut and poplar densities as practical reference points: materials above walnut density (approximately 630 kg/m³) are labeled "pass," materials between poplar and walnut (approximately 470-630 kg/m³) are labeled "OK," and materials at or below poplar are labeled "fail." These categories indicate likely durability concerns, not final acoustic suitability.

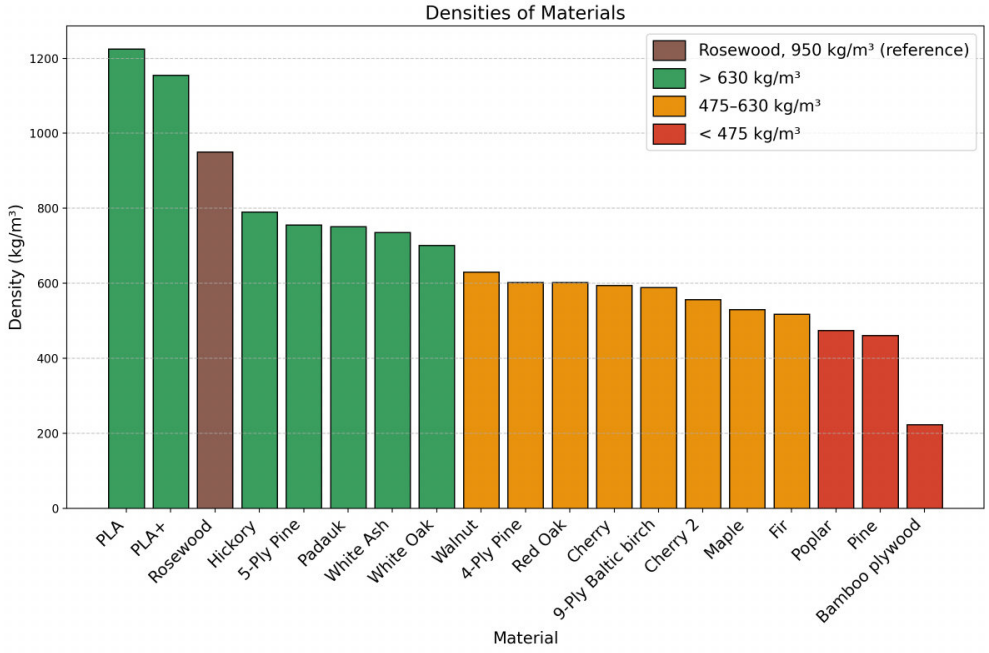


Figure 4. Densities of tested materials. Descriptive density categories use measured walnut and poplar values as practical reference thresholds.

A lower density does not necessarily make a material unusable. Softer materials may crack faster, but they can be easier to shape. Dense woods such as padauk and rosewood can damage or quickly wear woodworking equipment [16], [26]. A lower-density material might therefore still be valuable in educational contexts if it is inexpensive, easy to fabricate, and acceptable for practice, even if it is less durable than rosewood.

E/ρ ratio and damping

Figures 5 and 6 compare E/ρ ratio and damping, respectively. For a marimba bar of a specific size and shape, the E/ρ ratio determines the fundamental pitch, so this ratio is critical for ensuring that practice on non-rosewood bars transfers to rosewood instruments. Figure 5 shows that hickory, maple, 5-ply pine plywood, both cherry samples, walnut, and 4-ply pine plywood fell within the ±20% E/ρ range (0.0135–0.0202 (GPa·m³)/kg), along with the rosewood reference and the padauk control. Red oak, white ash, fir , white oak, pine, poplar, 9-ply Baltic birch plywood, PLA, and PLA+ fell outside the range. Although the PLA samples had high density, their low Young's modulus values made their pitch behavior diverge sharply from rosewood.

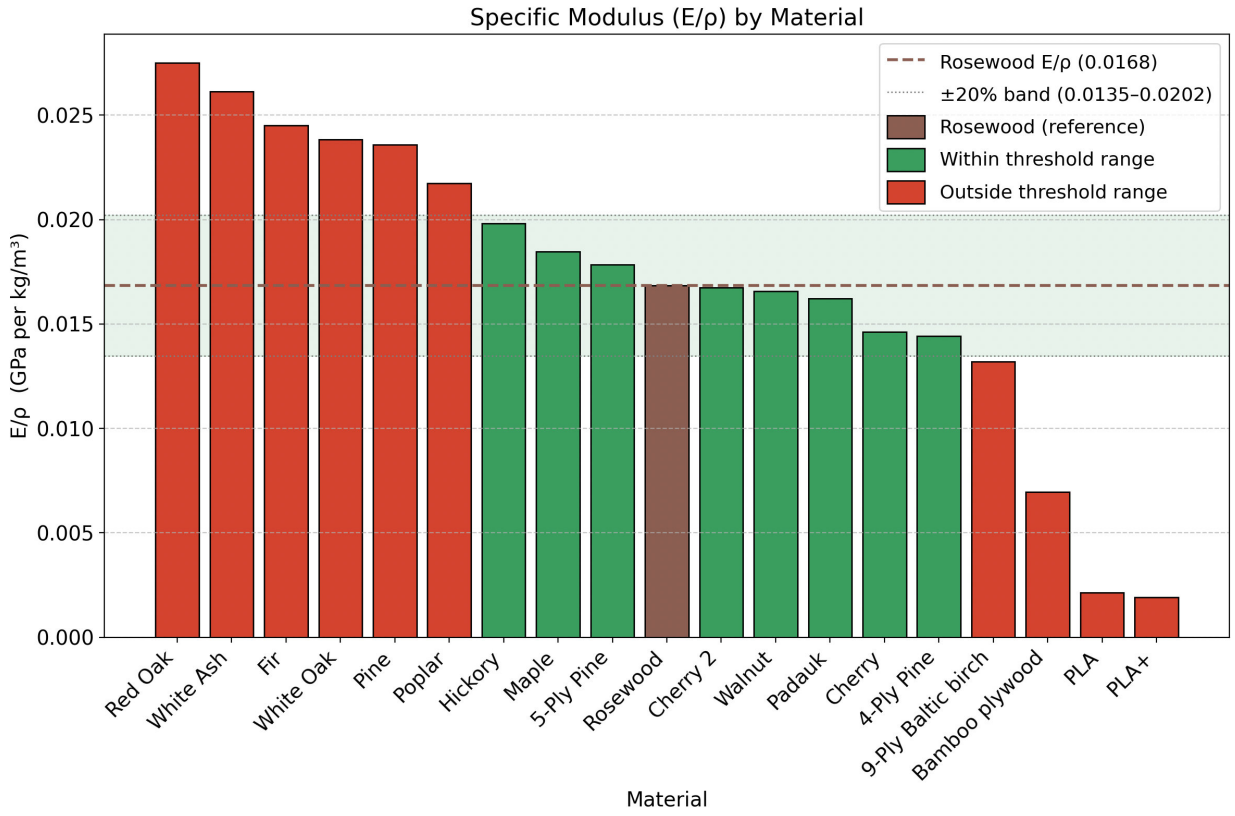


Figure 5. Young's-modulus-to-density ratio of tested materials compared with the ±20% rosewood-centered target range.

Figure 6 summarizes the damping results. No tested non-control material met the original rosewood-centered damping criterion; only rosewood and padauk fell within ±75% of rosewood's loss factor (0.0025–0.0177). As an exploratory second-stage screen, a new ±75% margin was then applied around padauk's damping value (0.0042–0.0294). Under this second padauk-centered stage, fir, hickory, and cherry were closest in damping. Fir was excluded because it failed the E/ρ criterion and had low density, leaving hickory and cherry as the strongest American wood candidates for future tuned-bar testing.

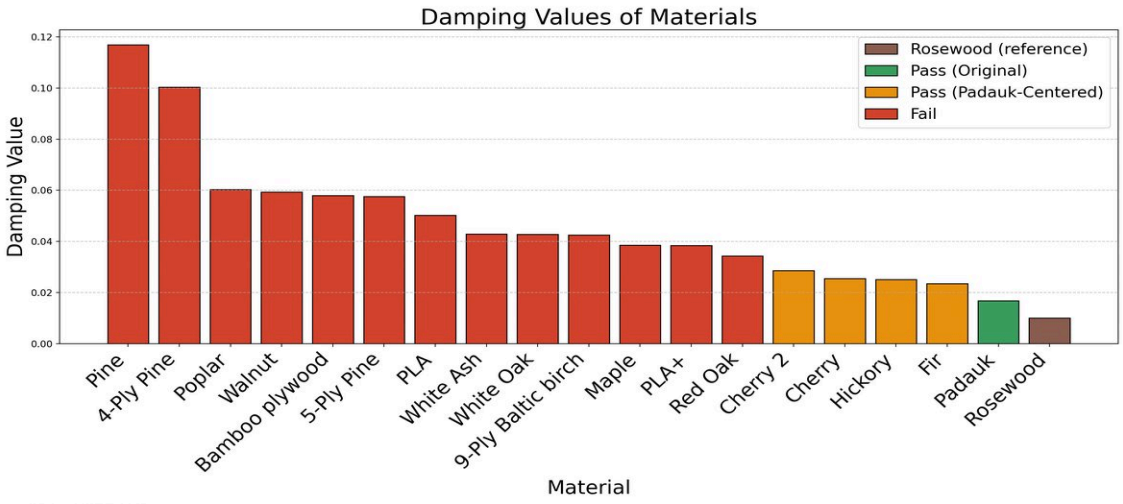


Figure 6. Loss factor values of tested materials compared with rosewood and padauk benchmarks.

Table 2. Condensed interpretation of selected tested materials.Padauk exclude as a control material.

Material	Screening result	Main limitation	Next step
Hickory	Strongest American wood candidate	Damping remained above the rosewood-centered criterion	Fabricate tuned bars; retest damping and durability
Cherry	Secondary candidate	Lower density and stiffness than rosewood	Test lighter-duty practice bars
PLA and PLA+	Not suitable as tested	Low Young's modulus relative to density	Only revisit with modified polymer designs
Bamboo plywood	Future research candidate	Tested sample was a glued approximation and failed target criteria	Retest commercial bamboo stock

The density, stiffness, and damping results together show why no single material property is sufficient. PLA and PLA+ were dense, which initially suggested possible durability, but their low Young's modulus values made them unsuitable as tested. Conversely, hickory stood out because its density, Young's modulus, and E/ρ ratio were comparatively close to rosewood, even though its damping did not meet the original rosewood-centered criterion. Cherry was less dense and less stiff than rosewood, but its E/ρ ratio remained close enough to justify further testing.

The combined screen therefore narrows, rather than solves, the material-selection problem. Hickory's main weakness is damping, not stiffness or density. Cherry's weaknesses are density and damping, but its E/ρ ratio is encouraging. The plastics' weakness is stiffness relative to density. The plywoods' weakness is that engineered construction did not, in the tested samples, produce a rosewood-like combination of stiffness, density, and damping.

DISCUSSION

Screening interpretation and limitations

These findings are best interpreted as a screening study rather than a finished instrument-design study. A real marimba bar is not simply a rectangular block: artisans tune bars by removing material from specific locations, shaping the underside, and balancing the fundamental with desired overtones. The present work therefore narrows the field before that more labor-intensive tuned-bar stage. Several methodological limitations remain. The supports were placed near the first bending-mode nodal region, but the exact nodal points were not measured; density variation and internal structure may shift the nodes and add slight damping from the twine supports. Future research could use laser vibrometry to identify modal shapes and nodal points more precisely [23].

The food scale and ruler introduced potential error in density and subsequent Young's modulus calculations. The ±0.5 g mass precision and ±1 mm dimensional precision were acceptable for a budget-constrained exploratory study, but they limit absolute accuracy. Audacity's inability to select decimal decibel values for half-width bandwidth measurement likely affected loss-factor calculations and could lead to overestimation or underestimation of damping. Additional uncertainty came from n = 1 sampling for most materials, unmeasured moisture content and grain orientation, room noise, microphone placement, and the lack of calibrated impact force. The two cherry samples had different density, Young's modulus, and damping values, indicating that the data represent individual samples and may not fully represent each species as a whole.

Material interpretation

No tested material besides rosewood and padauk fell within the original selection criteria. Rosewood was guaranteed to meet the criteria because its values defined the selection range. Padauk was not guaranteed to pass, but it was expected to do so as a positive control. Hickory and cherry nevertheless have properties that make them appropriate candidates for future marimba-bar experiments.

Cherry is less promising than hickory because it is significantly less dense than rosewood, averaging about 575 kg/m^3 compared with rosewood's 950 kg/m^3. Its Young's modulus also differs substantially from rosewood's, with an average near 9 GPa compared with rosewood's 16 GPa. However, cherry's E/ρ ratio remains close to rosewood's at approximately 0.0157 $(GPa*m^3)/kg$ compared with 0.0168 $(GPa*m^3)/kg$. Its damping value, approximately 0.0265, is higher than rosewood's 0.0101 but closer than many other tested materials. Cherry therefore represents a secondary candidate: its low density may create durability issues, but it may still be useful in lighter-duty educational or practice contexts.

Among the tested materials, hickory appears to be the strongest candidate for future testing. Its density, about 790 kg/m^3, falls between rosewood and padauk, and its Young's modulus differs from rosewood by only about 0.335 GPa. Because hickory also remains within the ±20% E/ρ target range, its pitch-producing geometry may be more transferable to rosewood than that of most tested materials. Its main unresolved issue is damping, which should be tested again using tuned bars and more precise spectral tools.

Impact and next steps

The findings could increase marimba accessibility by lowering bar-material costs. As of April 18th, 2025, retail listings for Honduran rosewood were listed at $53.00 per board foot (12"x12"x1") [27], while hickory was listed at $8.00 per board foot [28] and cherry at $6.20 per board foot [29]. Board-foot price is only one component of final instrument cost, but this difference is large enough to justify further study of whether domestic woods could support lower-cost practice instruments.

Future work should average data across larger numbers of samples for species-level accuracy and should fabricate tuned hickory and cherry bars for direct comparison with rosewood and padauk bars of the same pitch. Those bars should be evaluated for sustain time, overtone structure, perceived warmth, projection, mallet response, and durability after repeated strikes. Blind listening and playing tests with percussionists could separate measurable acoustic differences from player expectations about wood species. Hickory also warrants additional investigation into loss-factor reduction through thermal or mechanical modification, as modification methods have been explored for related xylophone-substitute research [9]. Bamboo deserves further study because its rapid growth and existing use in traditional Vietnamese t'rung instruments and tangential examples of Western-style bamboo xylophones show that there is a possibility of musical potential although the tested glued plywood sample did not meet the criteria [30], [31]. Finally, analyzing existing synthetic marimba bars, such as Yamaha's fiberglass-reinforced Acoustalon [17] and patented synthetic or hybrid marimba bars [32], could identify cost-effective and sustainable synthetic pathways.

Environmental context should remain central. Replacing Honduran rosewood with another scarce imported wood would fail the broader sustainability goal even if the acoustic match were excellent. American species such as hickory and cherry are attractive because they are more accessible domestically, but responsible sourcing, forestry practices, and supply consistency would still matter. A future cost model should include not only board-foot price but also machining time, rejected bars, tool wear, bar replacement rate, and the labor required to tune each material.

The study also suggests a useful division between performance instruments and practice instruments. A professional concert marimba may require the closest possible match to rosewood. A school or home-practice instrument may instead prioritize affordability, tuning stability, and adequate tactile transfer. Hickory may be most appropriate for the second category unless future damping modification improves its sustain. Cherry may be useful for lighter-duty practice contexts, especially if lower-density bars can be protected through mallet choice, bar geometry, or replaceable modular construction.

CONCLUSION

This study evaluated commercially available American woods, plywoods, and 3D-printed plastics as potential marimba-bar alternatives to Honduran rosewood. No tested material besides rosewood and padauk fully met the original selection criteria. However, hickory and cherry demonstrated enough promise to justify further investigation. Hickory is the strongest candidate because its density, Young's modulus, and E/ρ ratio are comparatively close to rosewood, while its primary weakness is damping. Cherry is less durable but remains interesting because its E/ρ ratio is close and its damping is nearer to the desired range than many other materials. Neither material should be described as a direct replacement yet; rather, both offer a foundation for more accessible and environmentally responsible

practice-instrument research. Future research should test larger sample sets, refine damping measurements, evaluate tuned bar geometry, and explore modification methods that could bring promising American woods closer to rosewood's acoustic behavior.

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